Sorting Shapes by What Makes Them That Shape
Answer Key

Kindergarten–Grade 1

Quick Answers

1. —	4. —	7. 6, 4, 3, 0	10. —
2. —	5. 3, 4, 6	8. —	
3. —	6. —	9. —	

Curriculum

Level: Kindergarten–Grade 1
K.G / 1.G – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

1. Count the straight sides of shape A (square) and shape B (triangle). Circle the shape with MORE sides.

Solution. Have the child touch each side once while counting. Shape A is a square: four straight sides. Shape B is a triangle: three straight sides. Four is more than three, so A gets circled. If the child counts corners instead of sides they still get the same answer here — ask them which one they counted, because problem 2 separates the two.

$$\boxed{A : 4, \quad B : 3, \quad 4 > 3}$$

2. Count the corners of shape C (rectangle) and shape D (hexagon). Circle the shape with FEWER corners.

Solution. A corner is where two sides meet. Shape C has four corners; shape D has six. Four is fewer than six, so C gets circled. A common slip is stopping at five on the hexagon — have the child mark each corner with a pencil dot as they count.

$$\boxed{C : 4, \quad D : 6, \quad 4 < 6}$$

3. Shapes E and F are both triangles, one small and one large. Milo says F is not a triangle because it is too big. Is Milo right?

Solution. Both shapes have exactly three straight sides and three corners, so the counts are equal. Size is not part of the rule for being a triangle — it is a non-defining attribute, like colour. The answer to circle is NO, and the reason to look for in the child's writing is some version of "it still has three sides". Accept any wording that names sides or corners; do not accept "because it looks like a triangle".

$$\boxed{E : 3, \quad F : 3, \quad 3 = 3}$$

4. Circle every shape with exactly FOUR straight sides. How many did you circle, and how many did you leave?

Solution. Circled (four straight sides): the square, the long rectangle, the trapezoid, and the rhombus — four shapes. Left uncircled: the triangle (three sides), the circle (no straight sides), and the hexagon (six sides) — three shapes. Four circled is more than three left, so the bigger number to circle is the four. Note that the rhombus and the trapezoid both count: they are not squares, but the rule was "four straight sides", not "looks like a square".

$$\boxed{4 > 3}$$

5. Count the corners of G (triangle), H (hexagon), and I (square). Write the numbers from fewest to most.

Solution. G has three corners, H has six, I has four. Sorting those three numbers from smallest to largest puts G first, then I, then H. Reading the shapes left to right gives 3, 6, 4 — the child has to reorder rather than copy, which is the point of the problem.

$$\boxed{3,\ 4,\ 6}$$

6. L is a square and M is the same square turned onto its point. Count the sides of each. Is the friend right that M is not a square?

Solution. Both have four straight sides and four corners, so the counts are equal and M is still a square. Turning a shape (its orientation) is a non-defining attribute — it changes nothing you can count. The answer to circle is NO. Many children this age name the turned square a "diamond"; that is the misconception this problem targets, so it is worth saying out loud that a diamond standing on its point with four equal straight sides is a square.

$$\boxed{L:4, \quad M:4, \quad 4=4}$$

7. Count the straight sides of the hexagon, square, triangle, and circle. Write the numbers from most to fewest.

Solution. Hexagon six, square four, triangle three, circle none — a circle has one curved edge and zero straight sides, which is exactly what separates it from the others. From most to fewest the numbers are 6, 4, 3, 0. Writing zero rather than leaving a blank is the part worth insisting on.

$$\boxed{6,\ 4,\ 3,\ 0}$$

8. Ana sorted the shapes into a top row and a bottom row. How many sides do the top-row shapes have? How many does the bottom-row triangle have? What was Ana's rule?

Solution. Every shape in the top row — the square, the rectangle, the trapezoid — has four straight sides. The triangle in the bottom row has three. Four is more than three, so the four gets circled. Ana's rule: the top row is "shapes with four straight sides". Accept "four sides" or "four corners"; both describe the same sort. Do NOT accept a rule based on size or on the shapes being "boxes", since the trapezoid is neither square nor rectangular.

$$\boxed{4 > 3}$$

9. Draw a four-sided, four-cornered shape that is NOT a square. Draw a shape with exactly three corners.

Solution. Left box — any closed figure with four straight sides that is not a square: a rectangle, a trapezoid, a rhombus, or an irregular quadrilateral all count. Check two things: the figure is closed (no gap), and it has four straight sides. Right box — any closed figure with three straight sides and three corners, at any size or turn. Marked by eye; the two things to look for are closure and straight edges.

10. Sam says a shape is a triangle because it is red. Say what is wrong with the rule, and write a better one.

Solution. What is wrong: colour is not part of what makes a shape a triangle. Red shapes can be circles, and triangles can be any colour — so Sam's rule would sort the shapes wrongly in both directions. A better rule: a shape is a triangle when it is closed and has three straight sides and three corners, no matter its colour, its size, or which way it is turned. Accept any child wording that names sides or corners as the test; the marker is whether the rule they write mentions something you can count.